

PHASE 1

Brainstorming & Ideation Phase

EduGenie - AI Powered Learning Assistant

Define the educational problem, target users, product idea, value proposition, initial feature set, and project scope.

Submission note: This document captures the project concept and scope; technical implementation evidence is documented in later phases.

Prepared as a phase-wise academic submission document based on the supplied EduGenie-Ai project archive.

1. Problem Statement

Students often need quick, personalized help with questions, explanations, practice quizzes, summaries, and learning paths. Traditional study materials are static, while a conversational AI assistant can generate context-aware learning support from a topic or passage supplied by the learner.

2. Proposed Idea

EduGenie is proposed as an AI-powered learning assistant that accepts student input and produces useful academic outputs through a simple web interface. The planned system uses Google Gemini for generative responses and FastAPI for the backend service.

Question Answering	Answer learner questions	Question text	Clear textual answer
Topic Explanation	Explain a concept	Topic text	Simplified explanation
Quiz Generator	Create practice questions	Topic/passage	MCQ-style quiz
Summarizer	Condense study material	Text passage	Short summary
Learning Recommendations	Suggest a learning sequence	Topic + level	Step-based learning path

3. Target Users

- School and college students who need on-demand study assistance.
- Self-learners preparing concepts before exams or interviews.
- Learners who prefer concise explanations and practice-oriented study support.

4. Core Value Proposition

One lightweight application can combine explanation, questioning, revision, practice, and guided learning in one place. The interface is designed to keep the learner interaction simple: choose a task, enter content, generate the result, and review or copy the response.

5. Initial Scope Decision

Web interface, AI generation, five learning tasks, basic validation, response display, copy action, health endpoint, environment-based configuration	Student account management, persistent profiles, payment, analytics dashboards, teacher administration, mobile-native clients, advanced document ingestion
--	--

6. Brainstorming Outcome

The ideation phase establishes EduGenie as a focused educational assistant rather than a general-purpose chatbot. The feature set is intentionally centered on academically useful transformations of learner-provided text or topics.

7. Concept Flow

Learner need -> Select learning task -> Enter question/topic/text -> Backend request -> Gemini generation -> Structured response -> Review / copy result

8. Success Indicators

- All five learning tasks are reachable from the user interface.
- Responses are displayed clearly and safely escaped in the browser.
- Quiz answers can provide immediate correctness feedback.
- A learner can copy the generated result for notes or revision.
- The application exposes a health endpoint for service verification.

9. Phase Conclusion

The brainstorming phase converts the educational problem into a concrete product concept with a manageable first release scope. Later phases translate this concept into requirements, design, implementation, testing, documentation, and demonstration artifacts.